Reconstruction of the Z Boson Mass Using CERN Open Data

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Abstract

Dimuon collision data from the CERN Open Data Portal were analysed to reconstruct the Z boson resonance using invariant mass reconstruction based on special relativity. Events within the expected Z boson mass region were modelled using a Gaussian signal and polynomial background. The fit yielded a reconstructed mass of (91.1004 ± 0.1263) GeV and a Gaussian width of (2.2081 ± 0.1507) GeV. The goodness-of-fit of the model was analysed using chi-squared statistics, giving a reduced chi-squared of 0.9623. The statistical significance of the reconstructed signal was calculated to be 12.3516σ, well above the standard discovery threshold of 5σ.

1 Introduction

The Z boson is an electrically neutral elementary particle that mediates the weak nuclear interaction. In high-energy particle collisions, Z bosons may be produced briefly before decaying into other particles, including muon–antimuon pairs. By analysing the four-momenta of these dimuon events, the invariant mass of the parent particle can be reconstructed using principles from special relativity. In this project, dimuon collision data from the CERN Open Data Portal are analysed to identify and reconstruct the Z boson resonance peak near 91 GeV [1]. The aim of this project is to reconstruct the Z boson mass spectrum and evaluate the statistical significance and quality of the reconstructed signal using computational fitting techniques.

2 Theory

2.1 Invariant Mass Reconstruction

In special relativity, the invariant mass is a quantity conserved in all inertial frames of reference [6]. It is derived from the conservation of four-momentum, defined as:

$$P^{\mu} = \left(\frac{E}{c}, \mathbf{p} \right), \tag{1}$$

where E is the total energy of the particle and $\mathbf{p}$ is its three-momentum vector.

The Lorentz-invariant magnitude of the four-momentum is given by:

$$P^{\mu} P_{\mu} = m_{\text{inv}}^2 c^2 = \frac{E^2}{c^2} - |\mathbf{p}|^2, \tag{2}$$

which leads to the invariant mass relation:

$$m_{\text{inv}}^2 c^4 = E^2 - |\mathbf{p}|^2 c^2. \tag{3}$$

For a particle decaying into two muons, the total four-momentum is obtained by summing the four-momenta of the decay products:

$$E = E_1 + E_2, \tag{4}$$

$$\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2. \tag{5}$$

The invariant mass of the parent particle can therefore be reconstructed from the measured muon energies and momenta:

$$m_{\text{inv}}^2 c^4 = (E_1 + E_2)^2 - |\mathbf{p}_1 + \mathbf{p}_2|^2 c^2. \tag{6}$$

In natural units, where $c = 1$, this simplifies to:

$$m_{\text{inv}}^2 = (E_1 + E_2)^2 - |\mathbf{p}_1 + \mathbf{p}_2|^2. \tag{7}$$

If the dimuon pair originates from the decay of a Z boson, the reconstructed invariant mass is expected to cluster around the accepted Z boson mass of approximately 91 GeV.

2.2 Resonance Peaks and Gaussian Distributions

When many dimuon collision events are analysed, most arise from background processes unrelated to Z boson production. However, events originating from genuine Z boson decays produce a concentration of reconstructed masses near the true particle mass. This produces a resonance peak in the invariant mass spectrum.

Experimentally, the resonance is broadened by detector resolution, finite statistics, and measurement uncertainties. The signal region can therefore be approximated using a Gaussian distribution of the form:

$$S(x) = A \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right], \tag{8}$$

where:

- A is the amplitude of the peak,
- μ is the reconstructed mean mass,
- σ is the Gaussian width.

Background collision events are approximated using a quadratic polynomial:

$$B(x) = a + bx + cx^2. \tag{9}$$

The total model fitted to the data is therefore:

$$f(x) = S(x) + B(x). \tag{10}$$

2.3 Chi-Squared Goodness-of-Fit

To evaluate how well the model describes the data, the chi-squared statistic is used:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i}, \tag{11}$$

where:

- O_i is the observed number of events in bin i,
- E_i is the expected number of events predicted by the model.

The reduced chi-squared is then calculated as:

$$\chi^2_\nu = \frac{\chi^2}{\nu}, \tag{12}$$

where ν is the number of degrees of freedom. A reduced chi-squared value close to 1 indicates good agreement between the model and the observed data.

2.4 Poisson Statistical Uncertainty

The number of detected events in each histogram bin follows approximately Poisson statistics. The statistical uncertainty in each bin is therefore estimated as

$$\sigma_i = \sqrt{N_i}, \tag{13}$$

where N_i is the number of observed events in bin i.

2.5 Statistical Significance

The statistical significance of the reconstructed signal is estimated using:

$$Z = \frac{S}{\sqrt{S+B}}, \tag{14}$$

where:

- S is the estimated number of signal events,
- B is the estimated number of background events.

Larger values of Z indicate a lower probability that the observed peak arose from statistical fluctuations alone. In particle physics, a significance of 5σ is commonly used as the threshold for a discovery-level observation.

3 Methodology

The program begins by loading dimuon collision data from the CERN Open Data Portal [1]. The dataset contains the energies and momentum components of two muons from each collision event: E_1, E_2, p_{x1}, p_{y1}, p_{z1}, p_{x2}, p_{y2}, and p_{z2}. These quantities are stored in NumPy arrays and used to reconstruct the invariant mass of each event using the relativistic invariant mass relation.

The reconstructed invariant mass spectrum is then filtered to include only events within the expected Z boson mass region between 70 GeV and 110 GeV. The selected events are histogrammed in order to visualise the distribution of reconstructed masses and prepare the data for statistical fitting.

The signal component, represented by a Gaussian distribution $S(x)$, and the background component, represented by a quadratic polynomial $B(x)$, are defined separately and combined into a total signal-plus-background model

$f(x)$. The model parameters are fitted to the histogrammed data using the `curve_fit()` function from SciPy, which performs non-linear least-squares optimisation by using estimates from the histogram as initial parameters before applying numerical optimisation. The covariance matrix returned by the fit is used to estimate the uncertainties in the fitted parameters [3, 4, 5].

The fitted model, signal component, and background component are then plotted together with the histogrammed data. Poisson error bars of magnitude $\sqrt{N}$ are included for each histogram bin, where N is the number of events in that bin. Residuals, defined as the difference between the observed data and the fitted model, are also plotted to assess the quality of the fit. Residuals fluctuating randomly about zero indicate reasonable agreement between the model and the observed data.

Finally, the goodness-of-fit of the model is evaluated using the reduced chi-squared statistic. The statistical significance of the reconstructed signal Z is then calculated using the estimated numbers of signal and background events within $\pm 2\sigma$ of the fitted peak position.

4 Results

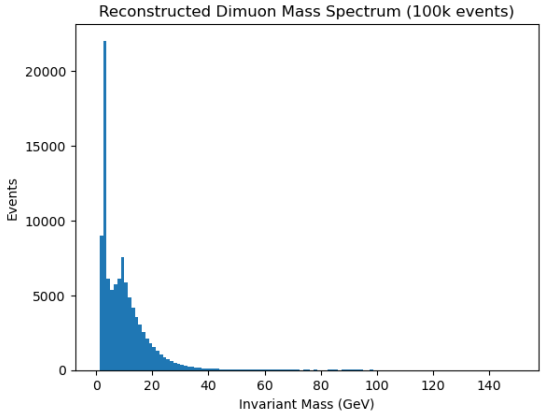


Figure 1: Reconstructed dimuon invariant mass spectrum for 100,000 collision events from the CERN Open Data Portal. The dominant structures below 40 GeV are consistent with the J/ψ (≈ 3.1 GeV) and Υ (≈ 9.46 GeV) meson resonances. A small excess of events is visible in the 70–110 GeV region, corresponding to Z boson candidates, which is examined in detail in Figure 2.

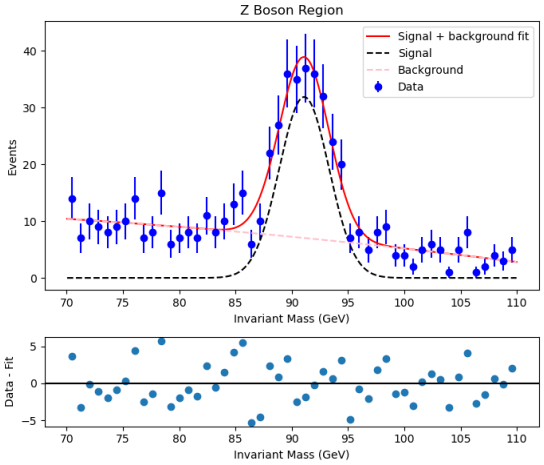


Figure 2: Upper panel: Invariant mass distribution of dimuon events within the Z boson signal region (70–110 GeV), with Poisson error bars. The red curve shows the combined signal-plus-background fit, decomposed into a Gaussian signal component (black dashed) and quadratic background (pink dashed). The fitted Gaussian mean is (91.1004 ± 0.1263) GeV with a width of (2.2081 ± 0.1507) GeV. Lower panel: Residuals (data minus fit) as a function of invariant mass, showing no systematic trends and confirming good agreement between the model and data ($\chi^2/\nu = 0.9623$).

4.1 Invariant Mass Spectrum

The reconstructed invariant mass spectrum across all 100,000 dimuon events is shown in Figure 1. The distribution strongly peaks at low invariant masses, dominant below 40 GeV. The sharp peak near 3 GeV and the broader enhancement around 9–10 GeV are consistent with the known resonances of the J/ψ ($\approx$ 3.1 GeV) and Υ ($\approx$ 9.46 GeV) mesons respectively. The rapid fall-off at higher masses reflects the steeply falling production cross-section for high-mass dimuon pairs.

A small but visible excess of events is observed in the 70–110 GeV signal region, corresponding to the expected Z boson mass of $\approx$ 91 GeV: this is examined in detail in the next subsection.

4.2 Z Boson Resonance Fit

Figure 2 shows the invariant mass distribution within the selected signal region (70–110 GeV), revealing a clear resonance peak consistent with the expected Z boson mass of 91.2 GeV. The data are well-described by the combined signal-plus-background model $f(x)$, in which a Gaussian component describes the signal and a quadratic polynomial accounts for background noise from other dimuon processes.

The fitted Gaussian mean, corresponding to the reconstructed Z boson mass, is (91.1004 ± 0.1263) GeV: this is a discrepancy of less than 0.1 GeV with the PDG value of 91.1876 GeV [2]. The Gaussian width is (2.2081 ± 0.1507) GeV,

giving a full width at half maximum (FWHM) of 5.2002 GeV. This is broader than the natural Z boson decay width of 2.495 GeV [2], which is expected as the measured width includes effects such as detector resolution, fitting approximations and binning effects.

The background contribution, visible as the pink dashed curve, is small relative to the signal across the peak region, suggesting the signal extraction is not very sensitive to the choice of background model $B(x)$.

4.3 Goodness-of-Fit

The lower panel of Figure 2 shows the residuals (differences between the observed data and fitted model), which fluctuate randomly about zero, indicating reasonable agreement between the model and the data and that no significant unmodelled contributions are present.

This is supported by the chi-squared statistic of $\chi^2 = 42.3420$ with 44 degrees of freedom (50 bins minus 6 free parameters), yielding a reduced chi-squared value of $\chi^2/\nu = 0.9623$. A reduced chi-squared value close to unity implies the model provides a good description of the observed invariant mass distribution, and the residual scatter is consistent with the expected statistical fluctuations from Poisson-distributed counts. Values significantly below 1 would suggest over-fitting or overestimated uncertainties, while values significantly above 1 would imply poor model agreement: the obtained value of 0.9623 therefore represents a good fit.

4.4 Statistical Significance

To quantify the strength of the signal, the number of signal events within 2σ of the fitted peak was estimated to be 209, while the number of background events was estimated to be 77. This suggests that, using the approximation:

$$Z = \frac{S}{\sqrt{S+B}}, \tag{15}$$

the statistical significance is found to be $Z = 12.3516\,\sigma$.

This value is substantially above the standard discovery threshold of 5σ, indicating the signal has a probability of less than 3×10^{-7} to arise from statistical fluctuations alone. The result therefore constitutes a clear observation of the Z boson resonance in this dataset. The relatively modest background contribution (approximately 27 percent of the total events in the observed region) further supports the robustness of this result, as the significance is not strongly dependent on the accuracy of the background model.

It should be noted that the significance estimator $Z = \frac{S}{\sqrt{S+B}}$ is a simplification, as it assumes Poisson statistics and does not account for systematic uncertainties (such as those arising from the choice of background parametrisation or detector effects) which would reduce the significance in a full experimental analysis. Nevertheless, for a dataset of this scale and with the background levels observed, the signal significance is unambiguous.

5 Discussion and Conclusion

This project successfully reconstructed the Z boson resonance from dimuon collision data from the CERN Open Data Portal using invariant mass reconstruction. The fitted Gaussian mean of $(91.1004 \pm 0.1263)\,\text{GeV}$ is in excellent agreement with the PDG value of $91.1876\,\text{GeV}$ [2], with a discrepancy of less than $0.1\,\text{GeV}$. This proves that relativistic invariant mass reconstruction from conservation of four-momentum is a robust method for identifying particle resonances.

The goodness-of-fit analysis yielded a reduced chi-squared of $\chi^2/\nu = 0.9623$, which confirms that the combined Gaussian signal and quadratic background model $f(x)$ provides a good description of the data. The statistical significance of $Z = 12.3516\,\sigma$ far exceeds the 5σ discovery threshold, essentially confirming the observed signal as a detection of the Z boson signal within the dataset.

There are also some limitations of this analysis worth highlighting. Firstly, the Gaussian approximation of the signal neglects the Breit–Wigner lineshape which more accurately describes a resonance decay. However, at the resolution and statistics of this dataset it has been shown that this approximation is sufficient (as confirmed by the chi-squared result). A Voigtian profile (a convolution of the Breit–Wigner and Gaussian distributions) would provide a more physically accurate model. Furthermore, the significance estimator:

$$Z = \frac{S}{\sqrt{S+B}}, \tag{16}$$

does not account for systematic uncertainties arising from fitting approximations, binning effects or detector effects. This would realistically reduce the significance.

References

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